Charlie Groll

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TECHNICAL SKILLS

Languages: Python · Swift

AI & Orchestration: watsonx Orchestrate · Multi-Agent Workflows

Backend & Cloud Exposure: FastAPI · WebSockets · AWS Cloud Fundamentals

Frameworks & Tools: SwiftUI · Firebase · Git · Xcode · MapKit · Core Location

EDUCATION

Queen's University — BASc, Computer Engineering — Expected May 2027

- Relevant Coursework: Operating Systems · Introduction to Data Science · Object-Oriented Programming · Software Development Methodology · Computer Networks
- Awards: Queen's Principal's Scholarship · TD Achieve the Dream Scholarship

EXPERIENCE

IBM Consulting — Student On Call (part-time) — Sep 2026 – Present
Developer Intern — Sep 2025 – Aug 2026

- Built and integrated a watsonx Orchestrate multi-agent intent router for a client chatbot, validating and classifying questions and handling ambiguity to produce structured inputs for knowledge-base retrieval.
- Built and deployed a human-in-the-loop, AI-assisted workflow that compares copied sales-dashboard data with previous input and generates formatted change reports used by IBM consultants.
- Prototyped a VS Code chatbot extension with code and file context, chat reset, and response-copying features and tested FastAPI and WebSocket approaches; separately improved file-upload handling and UI actions in another client tool.

TD Bank — Data Engineer Intern — May 2025 - Aug 2025

- Produced internal Power BI learning resources end-to-end by researching assigned topics, writing scripts and slides, and recording narrated demonstrations for TD employees.
- Explained each feature's purpose, relevant TD use cases, and live operation; revised materials based on manager and subject-matter-expert feedback before publishing to an employee-accessible internal SharePoint.

Queen's Engineering Society — Software Developer — Oct 2024 - Apr 2025

- Contributed research and planning for Student-BST, a student marketplace web app prototype, and implemented Firebase Authentication and Firestore integration.

Queen's University — Teaching Assistant — Sep 2023 - Apr 2024; Sep 2024 - Apr 2025

- Helped first-year engineering students debug code in weekly introductory programming labs and graded assignments.

PRODUCTS & PROJECTS

SweetSpots - Co-Founder & Sole iOS Developer · SwiftUI · Firebase · MapKit — Jul 2025 - Present

- Built, released, and continue to maintain a native iOS app for saving, organizing, mapping, and sharing places; reached 508 first-time downloads and 318 registered accounts as of Oct 2026.
- Implemented email, Google, and Apple authentication; real-time Firestore data models; map and list search; collections with cloud-hosted covers; place details; deep links; and an iOS share extension.
- Built geofence-based proximity reminders with multi-step location and notification permission handling, distance-prioritized monitoring within iOS's 20-region limit, and targeted unit tests.

Spotify Release Tracker - Python · Spotify Web API · Telegram Bot API — Jun 2026

- Built a Python automation tool that tracks releases for artists in a Spotify playlist and sends Telegram alerts for albums, singles, and guest features.
- Implemented Spotify OAuth, pagination, rate-limit handling, release deduplication, and persistent local JSON state; documented secure environment-variable setup in a public repository.